



ELK406E

Distribution of Electrical Energy

Prof.Dr. Mustafa BAĞRIYANIK

E-mail: bagriy@itu.edu.tr

Underground Distribution

- Where general appearance, economics, congestion, or maintenance conditions make overhead construction inadvisable, underground construction is specified.
- While overhead lines have been ordinarily considered to be less expensive and easier to maintain, developments in underground cables and construction practices have narrowed the cost gap to the point where such systems are competitive in urban and suburban residential installations, which constitute the bulk of the distribution systems



In return for the considerable expense, electric customers can receive a number of potential benefits from the undergrounding of their overhead systems. The following is a list of benefits most often mentioned in undergrounding reports and studies:

Potential Benefits of Underground Electric Facilities

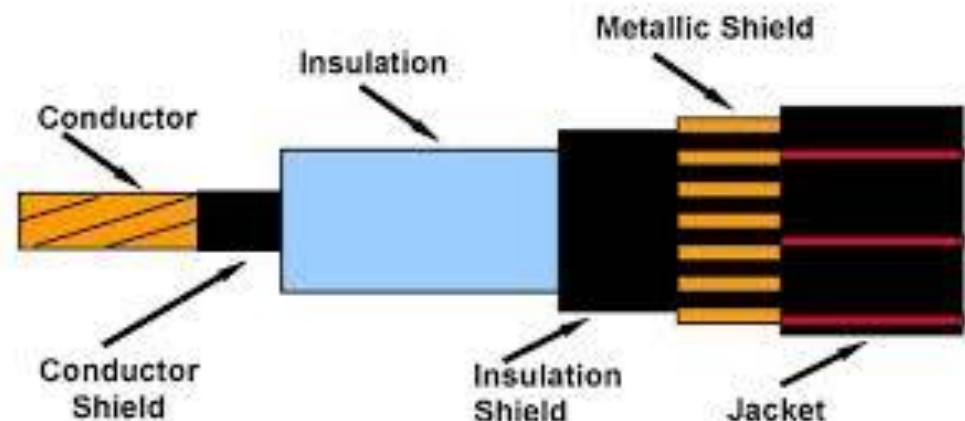
- Improved aesthetics;
- Lower tree trimming cost;
- Lower storm damage and restoration cost;
- Fewer motor vehicle accidents;
- Reduced live-wire contact;
- Fewer outages during normal weather;
- Far fewer momentary interruptions;
- Improved utility relations regarding tree trimming;
- Fewer structures impacting sidewalks.



Potential Disadvantages of Underground Electric Facilities

- Stranded asset cost for existing overhead facilities;
- Environmental damage including soil erosion, and disruption of ecologically-sensitive habitat;
- Utility employee work hazards during vault and manhole inspections;
- Increased exposure to dig-ins;
- Longer duration interruptions and more customers impacted per outage;
- Susceptibility to flooding, storm surges, and damage during post-storm cleanup;
- Reduced flexibility for both operations and system expansion;
- Reduced life expectancy
- Higher maintenance and operating costs;
- Higher cost for new data bandwidth.

Underground Cable Construction



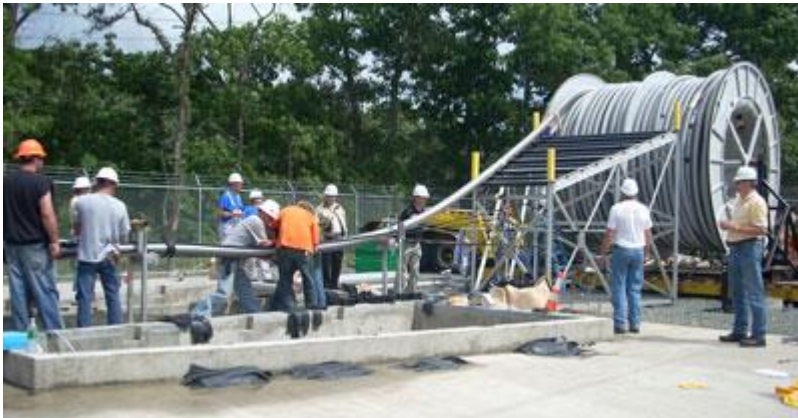
Underground construction is expensive, and costs vary widely.

The main factors that influence underground costs are:

- **Degree of development** — Roads, driveways, sidewalks, and water pipes — these and other obstacles slow construction and increase costs.
- **Soil condition** — Rocks and frozen ground increase overtime pay for cable crews.
- **Urban, suburban, or rural** — Urban construction is more difficult not only because of concrete, but also because of traffic. Rural construction is generally the least expensive per length, but lengths are long.
- **Conduit** — *Concrete-encased ducts* cost more than *direct-buried conduits*, which cost more than preassembled flexible conduit, which cost more than directly buried cable with no conduits.



- **Cable size and materials** — The actual cable cost is a relatively small part of many underground applications. A 1/0 aluminum full-neutral 220-mil TR-XLPE cable costs just under \$2 per ft; with a 500-kcmil conductor and a one-third neutral, the cable costs just under \$4 per ft.
- **Installation equipment** — Bigger machines and machines more appropriate for the surface and soil conditions ease installations.



Cables

- At the center of a cable is the phase conductor,
- then comes a semiconducting conductor shield,
- the insulation,
- a semi-conducting insulation shield,
- the neutral or shield, and
- finally a covering jacket.

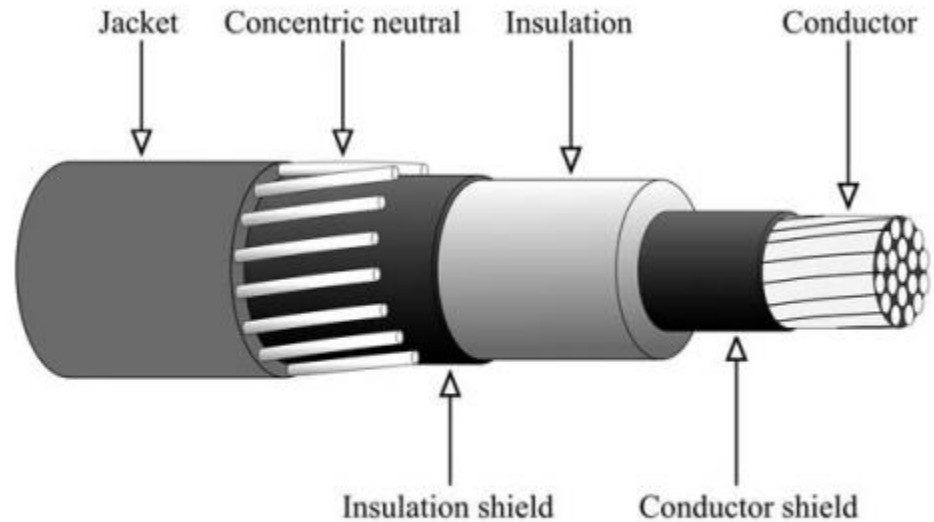


FIGURE 3.2

A concentric neutral cable, typically used for underground residential power delivery.

- Most distribution cables are single conductor.
- Two main types of cable are available: concentric-neutral cable and power cable.
- Concentric-neutral cable normally has an aluminum conductor, an extruded insulation, and a concentric neutral

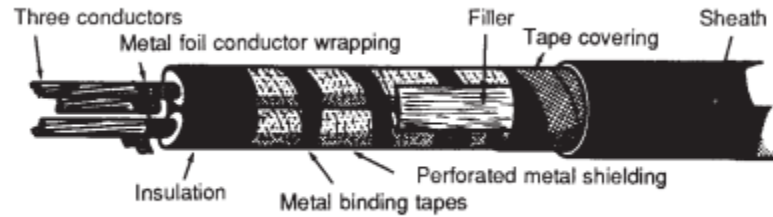


Figure 6-1. An underground cable.

- A concentric neutral is made from several copper wires wound concentrically around the insulation; the concentric neutral is a true neutral, meaning it can carry return current on a grounded system.
- Underground residential distribution normally has concentric-neutral cables; concentric-neutral cables are also used for three-phase mainline applications and three-phase power delivery to commercial and industrial customers.

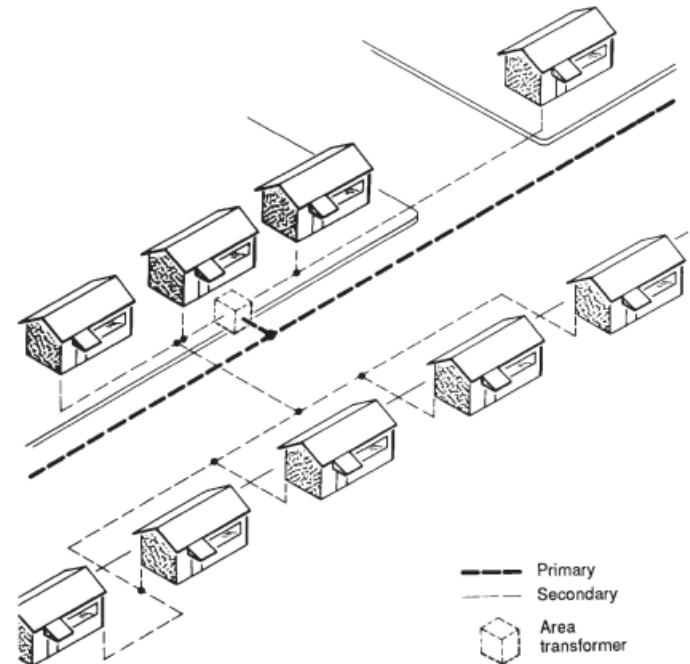


Figure 6-3. Underground residential layout using an area transformer.

- Power cable has a copper or aluminum phase conductor, an extruded insulation, and normally a thin copper tape shield.
- On utility distribution circuits, power cables are typically used for mainline feeder applications, network feeders, and other high current, three-phase applications.
- Many other types of medium-voltage cable are available.
- These are sometimes appropriate for distribution circuit application: three-conductor power cables, armored cables, aerial cables, fire-resistant cables, extra flexible cables, and submarine cables

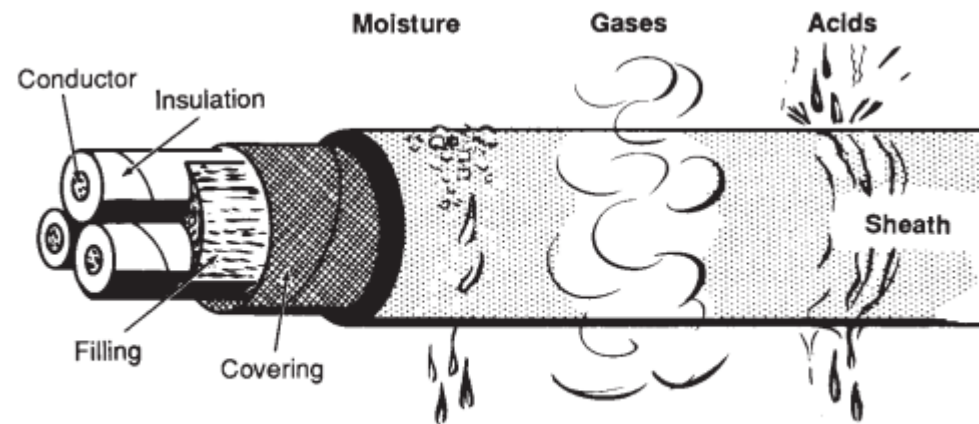
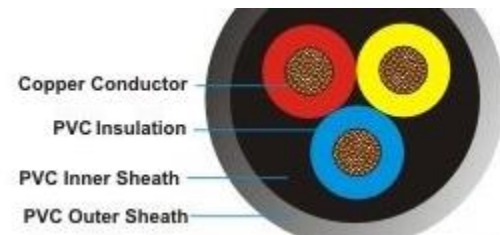


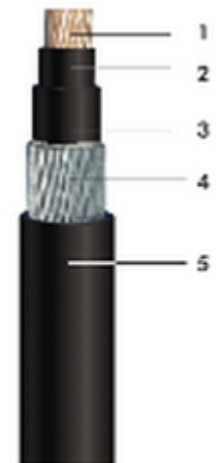
Figure 6-10. Underground cable protection provided by the sheath.

Jacket

- The jacket provides some (but not complete) protection against water entry.
- It also provides mechanical protection for the neutral.
- Several materials are used for jackets.
- Polyvinyl chloride (PVC) was one of the earliest jacketing materials and is still common
- PVC is used mainly in low voltage application because of its low insulation strength.
- XLPE - cross linked polyethylene.



1. Conductor: Copper
2. Insulation: XLPE
3. Inner sheath: PVC
4. Armour: Single layer Aluminium wire
5. Outer sheath: PVC



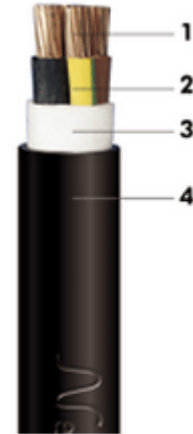
NYY Multi Core Cable

0.6/1 kV PVC insulated, multi core cables with copper conductor

Used as lighting and power distribution cable. Installed in areas where severe mechanical stress is avoided. Can be laid in doors, outdoors buried or in cable trenches



1. Conductor: Copper
2. Insulation: PVC
3. Filler: PVC
4. Outer Sheath: PVC

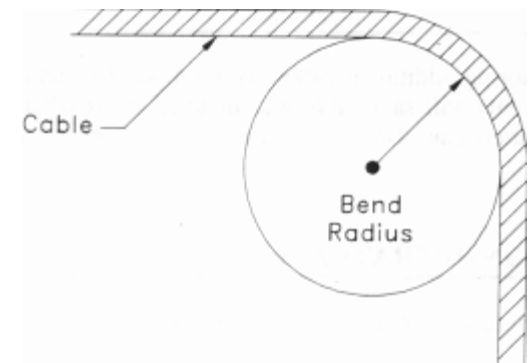
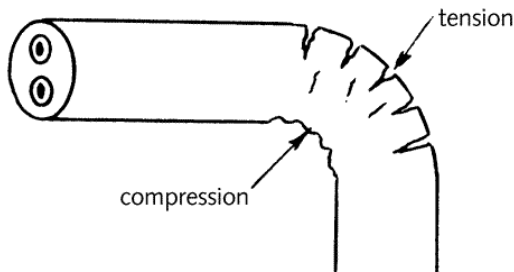


Power Cable; NYCY

As NYY, but with concentric outside conductor of bare copper wires, with counter spiral of copper band under the outer sheath

Don't overbend the cables

- A cable's minimum bending radius is directly related to its construction and directly affects cable insulation.
- Two values are used in defining a minimum bending radius: One describes the radius of a cable bend made by training a cable in its place after its installation; the other defines a bend a cable can safely negotiate without damage during the pulling process.
- As you would expect, the pulling radius is always greater than the training radius for the same cable because you expose the cable to greater physical stress during a pull.



D: the cable diameter

R: Bend Radius

For 3-conductors cable

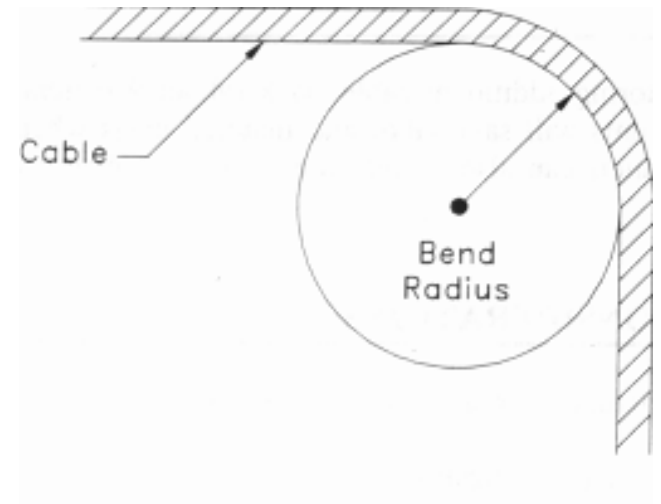
$R = 15 \times D$ (HV)

$R = 12 \times D$ (LV)

For single conductor cable

$R = 15 \times D$ (HV)

$R = 15 \times D$ (LV)



Pulling radius.

MV cables with overlapping tape shielding systems are particularly susceptible to damage if you don't use properly sized bends. If the radius is too small, the overlapping layer of tape shielding separates as the cable progresses through the bend. As the cable straightens out, the tape shielding buckles as it tries to come back together. In fact, it may actually cut into the cable's insulation. Since this cable construction usually features an overall jacket, you probably won't notice the damage, and the cable will have a reduced service life.

Training radius.

- Training is the positioning of cable when it is not under tension.
Bending is the positioning of cable when it is under tension.

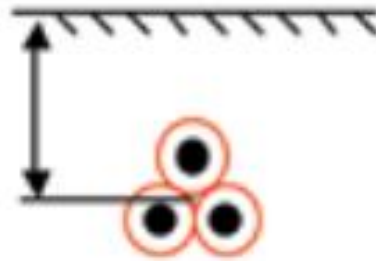
- IEC 60228 is the International Electrotechnical Commission's international standard on conductors of insulated cables.
- Among other things, it defines a set of standard wire cross-sectional areas:

International standard wire sizes (IEC 60228)					
0.5 mm ²	0.75 mm ²	1 mm ²	1.5 mm ²	2.5 mm ²	4 mm ²
6 mm ²	10 mm ²	16 mm ²	25 mm ²	35 mm ²	50 mm ²
70 mm ²	95 mm ²	120 mm ²	150 mm ²	185 mm ²	240 mm ²
300 mm ²	400 mm ²	500 mm ²	630 mm ²	800 mm ²	1000 mm ²

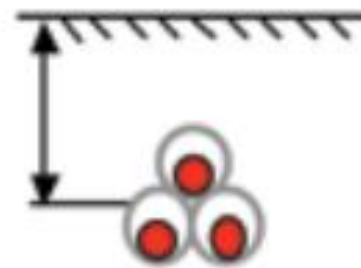
The following conditions have been assumed to calculate the current ratings:



Directly buried in ground



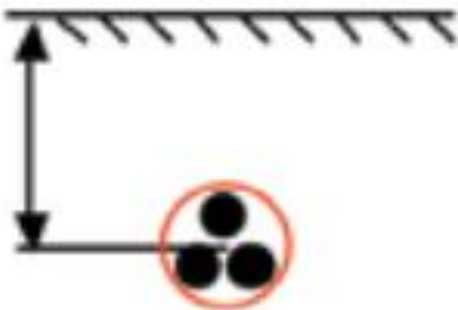
Laid in single way ducts



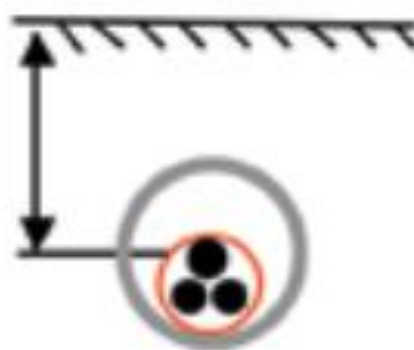
In air



Directly buried in ground



Laid in single way ducts



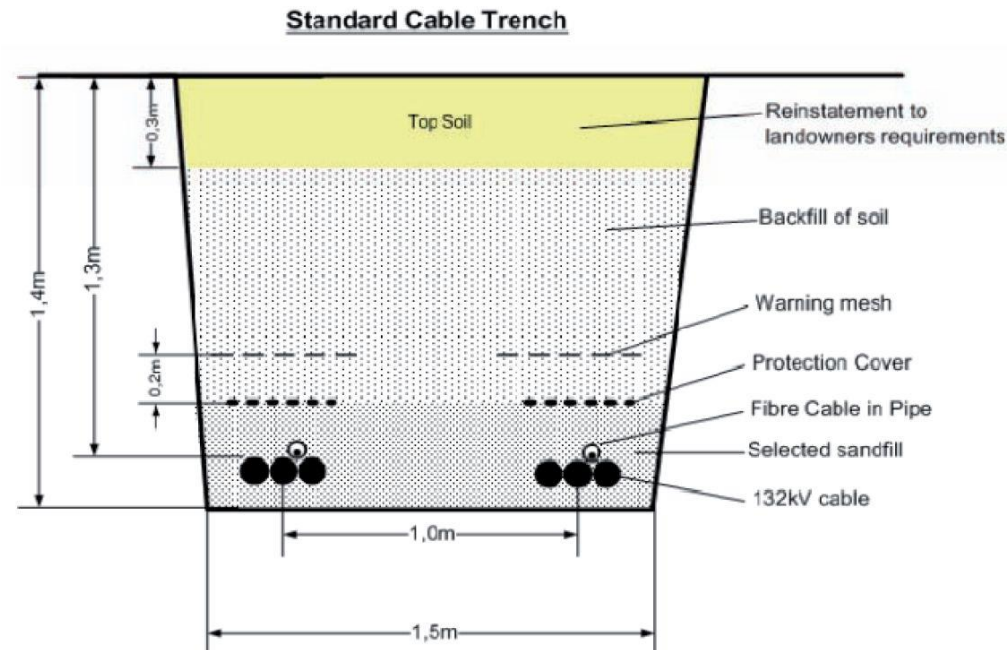
In air

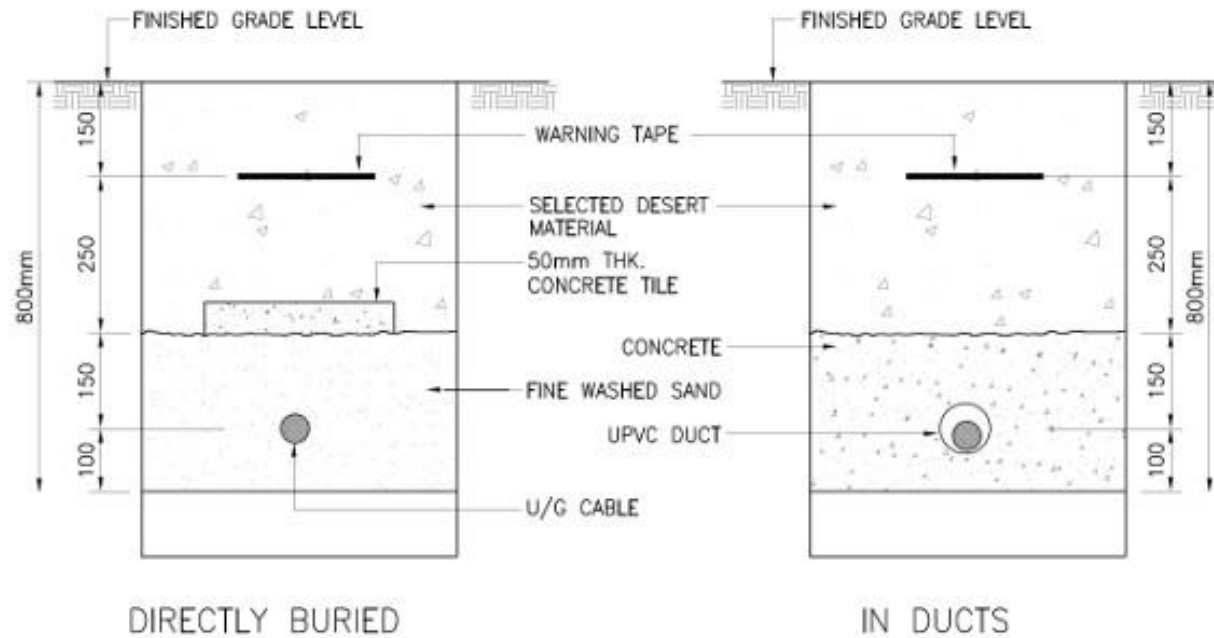


Cable Trench

Trenching — This is the most common way to install cables, either *direct-buried* or *cables in conduit*.

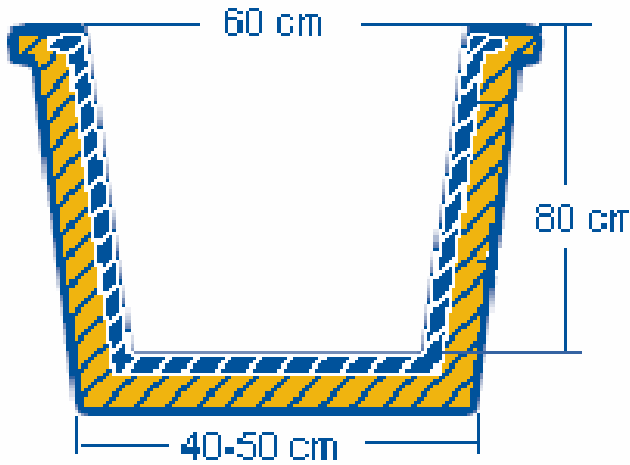
After a trench is dug, cable is installed, backfill is added and tamped, and the surface is restored.





NOTE: WHEN DIRECTLY BURIED CABLES ARE CROSSED WITH OTHER SERVICES THEN THEY SHOULD BE INSTALLED IN DUCT AND KEPT AT MINIMUM DISTANCE OF 300 mm FROM THE OTHER SERVICE. EXACT DETAILS FOR EACH CASE TO BE APPROVED BY SUPERVISING ENGINEER AT SITE.

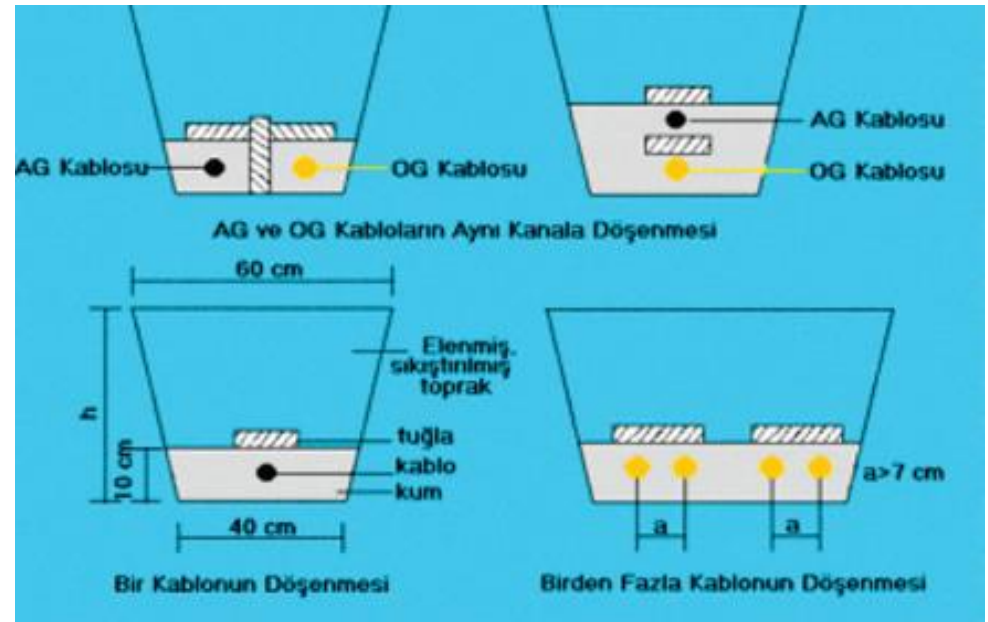
INSTALLATION OF UNDERGROUND CABLES



$U=1 \text{ kV}$ $h=40-50 \text{ cm}$

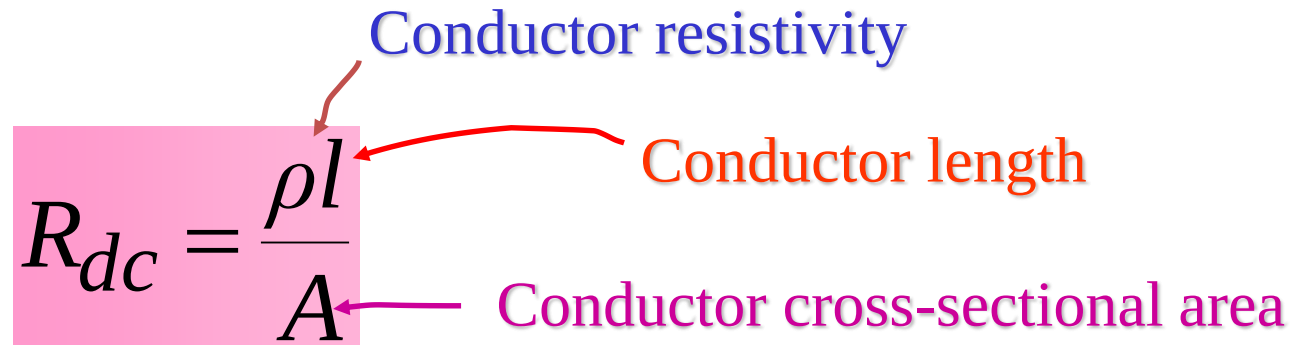
$U=1-10 \text{ kV}$ $h=60-70 \text{ cm}$

$U=10-35 \text{ kV}$ $h=80-100 \text{ cm}$



Line Resistance

The resistance of the conductor is very important in transmission efficiency evaluation and economic studies. The dc resistance of a solid round conductor at a specified temperature is

$$R_{dc} = \frac{\rho l}{A}$$


The diagram shows the formula $R_{dc} = \frac{\rho l}{A}$ with three labels and arrows pointing to the variables:

- Conductor resistivity** (blue text) points to ρ with a blue arrow.
- Conductor length** (orange text) points to l with an orange arrow.
- Conductor cross-sectional area** (purple text) points to A with a purple arrow.

The conductor resistance is best determined from **manufacturers' data**.

Resistance

- Cable conductor resistance is an important part of impedance that is used for fault studies and load flow studies.
- Resistance also greatly impacts a cable's ampacity.
- The major variable that affects resistance is the conductor's temperature; resistance rises with temperature.
- Magnetic fields from alternating currents also reduce a conductor's resistance relative to its dc resistance.
- At power frequencies, skin effect is only apparent for large conductors and proximity effect only occurs for conductors in very tight configurations.
- The starting point for resistance calculations is the dc resistance. From there, we can adjust for temperature and for frequency effects.

Resistance increases with temperature as

$$R_{t_2} = R_{t_1} \frac{M + t_2}{M + t_1}$$

where

R_{t_2} = resistance at temperature t_2 given, °C

R_{t_1} = resistance at temperature t_1 given, °C

M = a temperature coefficient for the given material

= 228.1 for aluminum

= 234.5 for soft-drawn copper

The ac resistance of a conductor is the dc resistance increased by a skin effect factor and a proximity effect factor

$$R = R_{dc} (1 + Y_{cs} + Y_{cp})$$

where

R_{dc} = dc resistance at the desired operating temperature, $\Omega/1000$ ft

Y_{cs} = skin-effect factor

Y_{cp} = proximity effect factor

Cables have significant **capacitance**, much more than overhead lines.

Conductor Sizing

- To find the proper conductor size,
 - the type of conductor (copper or aluminum),
 - the distance in km,
 - the current that are needed for the loadare must be known.
- There are an amazing variety of sizes and types of conductors.
- Several electrical, mechanical, and economic characteristics affect conductor selection

Conductor Sizing

The line conductor must satisfy several conditions at the same time in order to ensure a safe and reliable installation.

- It must carry the permanent full load current
- It must carry the short-time overcurrents
- It must not cause voltage drops likely to result in an inferior performance of certain loads
- It must not cause higher power losses
- It must have high tensile strength in order to withstand mechanical stresses

Ampacity — The peak current-carrying capability of a conductor limits the current (and power) carrying capability.

Economics — Often we will use a conductor that normally operates well below its ampacity rating.

- The cost of the extra aluminum pays for itself with lower I^2R losses; the conductor runs cooler.
- This also leaves room for expansion.

Mechanical strength — Especially on rural lines with long span lengths, mechanical strength plays an important role in size and type of conductor.

- Stronger conductors like ACSR are used more often.
- Ice and wind loadings must be considered.

Corrosion — While not usually a problem, corrosion sometimes limits certain types of conductors in certain applications

Ampacity

- A cable's ampacity is the maximum continuous current rating of the cable.
- We should realize that while we may derive one number, say 480 A, for ampacity during normal operations for a given conductor, there is nothing magic about 480 A. The cable will not burst into flames at 481 A; the 480 A is simply a design number. We don't want to exceed that current during normal operations.
- The insulation temperature is normally the limiting factor. By operating below the ampacity of a given cable, we keep the cable insulation below its recommended maximum temperature. Cross-linked polyethylene cables are rated for a maximum operating temperature of 90°C during normal operations.
- Operating cables above their ampacity increases the likelihood of premature failures: water trees may grow faster, thermal runaway-failures are more likely, and insulation strength may decrease.
- In addition to absolute temperature, thermal cycling also ages cable more quickly.

- Ampacity most often limits the loading on a cable; rarely, voltage drop or flicker limits loadings.
- Relative to overhead lines, cables of a given size have lower impedance and lower ampacities.
- So cable circuits are much less likely than overhead circuits to be voltage-drop limited.
- Only very long cable runs on circuits with low primary voltages are voltage-drop limited.

- Ampacity is not the only consideration for cable selection; losses and stocking considerations should also factor into cable selection.
- Choosing the smallest cable that meets ampacity requirements has the lowest initial cost, but since the cable is running hotter, the cost over its life may not be optimal because of the losses.
- Also allow for load growth when selecting cables.

The variables that most impact ampacity

- Sheath resistance
- Spacings
- Conductor temperature
- Loss factor
- Conduits
- Soil thermal resistivity and temperature
- Earth interface temperature
- Current unbalance
- System voltage and insulation thickness
- Number of cables
- Cable crossings and other hotspots

Fault Withstand Capability

- Short-circuit currents through a conductor's resistance generates tremendous heat.
- All cable between the source and the fault is subjected to the same phase current.
- For cables, the weakest link is the insulation; XLPE have a short-duration upper temperature limit of 250°C.
- The shortcircuit current injects energy as a function of the fault duration multiplied by the square of the current.

Damage to the shield or the neutral is more likely than damage to the phase conductor. During a ground fault, the sheath may conduct almost as much current as the phase conductor, and the sheath is normally smaller.

- In the vicinity of the fault, the fault current can cause considerably more damage to the shield or neutral.
- With a concentric neutral, the fault current may only flow on a few strands of the conductor until the cable has a grounding point where the strands are tied together.
- **Excessive temperatures can damage the insulation shield, the insulation, and the jacket.**
- In addition, the temperature may reach levels that melt the neutral strands.
- A tape shield can suffer similar effects: where tape layers overlap, oxidation can build up between tape layers, which insulate the layers from each other. This can restrict the fault current to a smaller portion of the shield.
- Additionally, **where the fault arc attaches, the arc injects considerable heat into the shield or neutral, causing further damage at the failure point.** Some additional damage at the fault location must be tolerated, but the arc can burn one or more neutral strands several feet back toward the source.

Design based on economic considerations

The distribution costs can be expressed as the amount of Money needed to distribute 1 kWh

These costs consist of

- Costs independent of the distributed power or energy (amortization, depreciation, maintenance).
- Costs due to power and energy losses
 - **Power loss costs** result from the need to provide the network with sufficient power to compensate these losses. They do not depend on the line's annual utilization period
 - **Energy loss costs** are determined by the electric distribution operation and therefore depend both on the power as well as on the annual utilization period of the line.

Determination of total power demand

- use the population

$$P_p = N a$$

N , population

a , power consumption w/person

-add 10 % of P_p for street lighting (P_a)

-add individual loads (P_k)

$$P = P_p + P_a + \sum_{j=1}^{nk} P_{k_j}$$

- use the measurements and diversity factor

SERVICE FACTORS

- If all the lights, irons, washing machines, refrigerators, television sets, motors, and other equipment in all the consumers' premises were to be turned on at one time, it would be impossible for the utility company to supply the energy necessary to run all these appliances.
- For a company to install generators, transformers, transmission and distribution lines, and other equipment to provide for this total connected load would be extremely uneconomical since all this equipment will not be operating at the same time.
- Therefore, studies are made of the personal habits of people and the routines of households, offices, stores, and factories, to determine what probable load the utility company will have to supply at any one time

Figure 7-2 illustrates typical peak and off-peak load periods.

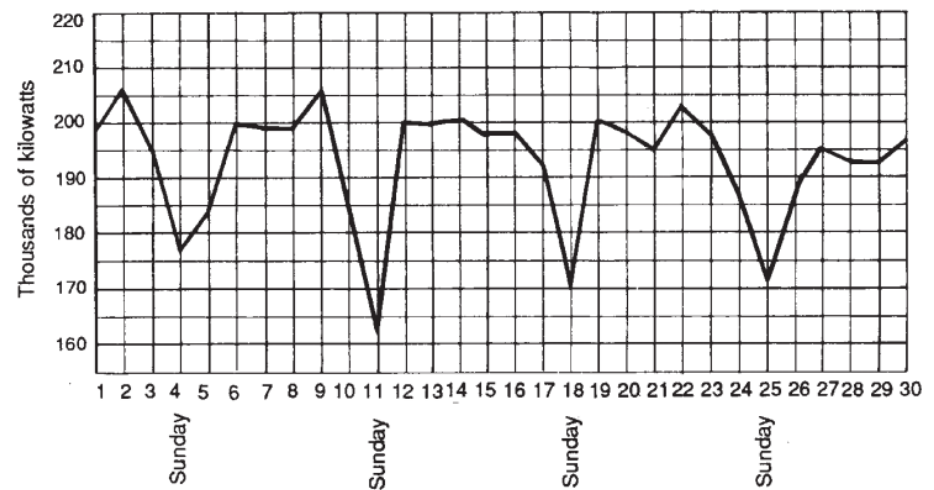


Figure 7-2. Daily peak load for a typical month. How many kilowatts a typical utility company supplies consumers in an average month. Note consistently low peaks on Sunday.

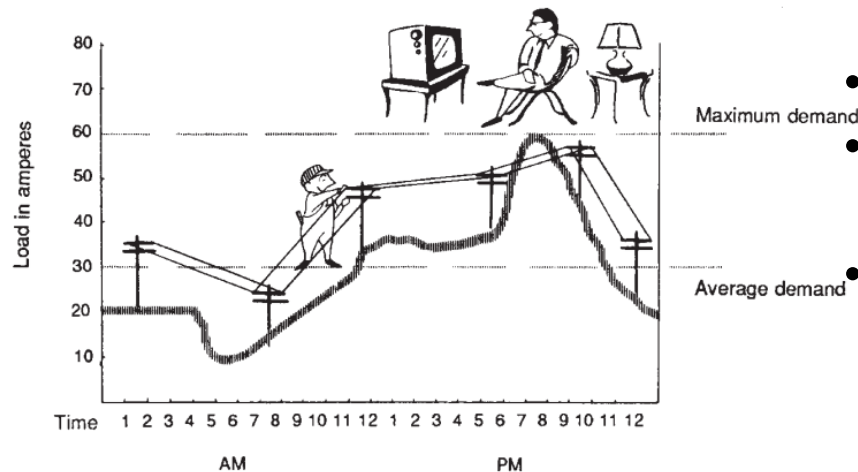


Figure 7-3. The utility company must be equipped to meet the consumer's maximum demand.

- *The amount of electricity a consumer uses at any given moment is called his “**demand.**”*
- It varies from hour to hour.
- *The highest value which the demand reaches is called the “**maximum demand**”*
- It is evident then that this maximum demand for electricity (and not the actual consumption of electricity) determines the size and type of equipment necessary to supply the consumer.

Load Factor

- If a consumer approaches this maximum demand often during the day, his or her consumption of electricity will be large.
- If the average demand in a given period of time is taken and compared to the maximum demand, the percentage is an estimate of the extent to which the facilities are used.
- This comparison, known as the **load factor** is actually the ratio between the average demand and the maximum demand in percent form.

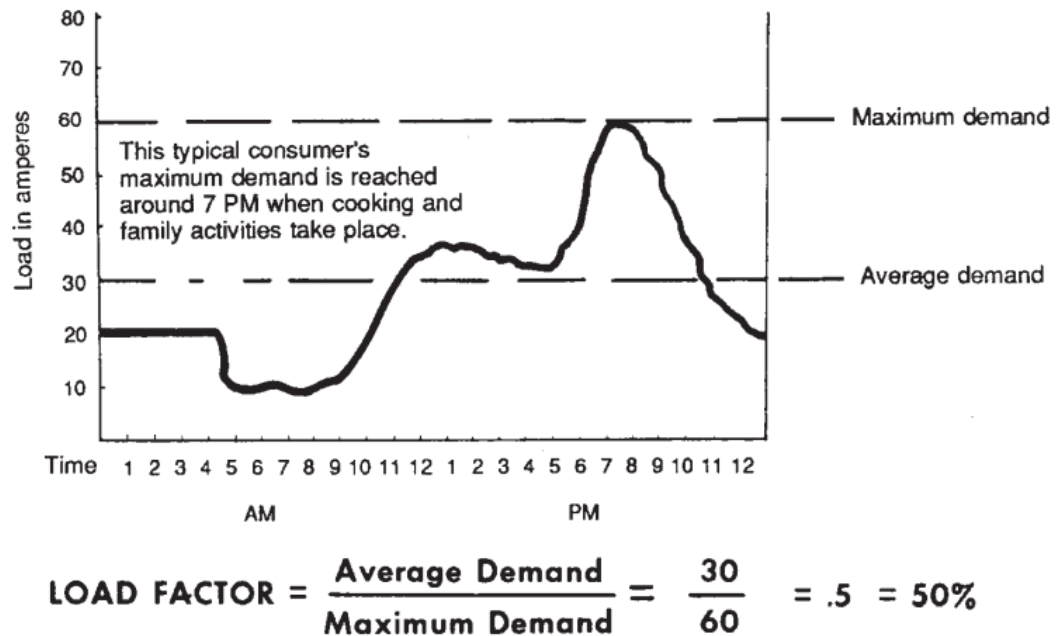


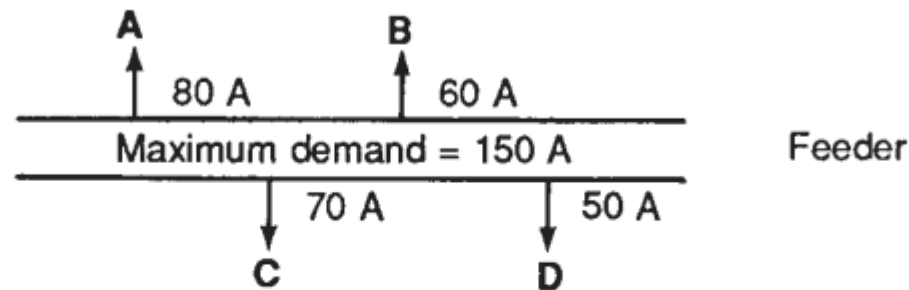
Figure 7-4. A consumer's load demand varies.

Diversity factor

- The time of the maximum demand of each consumer is important.
- Consider two houses with the same maximum demand but occurring at different times. If these two houses were fed from the same feeder, the demand on the feeder would be less than the sum of the two demands. This condition is known as diversity
- If we compare the actual maximum demand on the feeder with the sum of the maximum demands of all the consumers supplied by the feeder (regardless of when they occur), the relationship, expressed in percent, is known as the diversity factor

$$\text{diversity factor} = \frac{\text{Sum of individual max demand}}{\text{max demand on feeder}}$$

Four consumers being serviced by same feeder



Monday: A Reaches maximum demand of 80 A
Tuesday: B Reaches maximum demand of 60 A
Wednesday: C Reaches maximum demand of 70 A
Thursday: D Reaches maximum demand of 50 A

$$\text{Diversity factor} = \frac{\text{Sum of total demands}}{\text{Maximum demand on feeder}} = \frac{260}{150} = 1.73 \times 100\% = 173\%$$

Figure 7-8. Diversity factor.

Guide to Electrical Power Distribution Systems, Sixth Edition Anthony J. Pansini

Power loading of an installation

- The **installed power** is the sum of the nominal powers of all power-consuming devices in the installation.
- This is not the power to be actually supplied in practice.
- This is the case for electric motors, where the power rating refers to the output power at its driving shaft. The input power consumption will evidently be greater
- Fluorescent and discharge lamps associated with stabilizing ballasts, are other cases in which the nominal power indicated on the lamp is less than the power consumed by the lamp and its ballast.

For a power supply from a LV public-supply network, or through a MV/LV transformer, the significant quantity is the apparent power in kVA

Factor of maximum utilization (ku)

In normal operating conditions the power consumption of a load is sometimes less than that indicated as its nominal power rating, a fairly common occurrence that justifies the application of an utilization factor (ku) in the estimation of realistic values.

This factor must be applied to each individual load, with particular attention to electric motors, which are very rarely operated at full load.

In an industrial installation this factor may be estimated on an average at 0.75 for motors.

For incandescent-lighting loads, the factor always equals 1.

For socket-outlet circuits, the factors depend entirely on the type of appliances being supplied from the sockets concerned.

For Electric Vehicle the utilization factor will be systematically estimated to 1, as it takes a long time to load completely the batteries (several hours) and a dedicated circuit feeding the charging station or wall box will be required by standards.

Factor of simultaneity (ks)

It is a matter of common experience that the simultaneous operation of all installed loads of a given installation never occurs in practice, i.e. there is always some degree of diversity and this fact is taken into account for estimating purposes by the use of a simultaneity factor (ks).

The factor ks is applied to each group of loads (e.g. being supplied from a distribution or sub-distribution board). The determination of these factors is the responsibility of the designer, since it requires a detailed knowledge of the installation and the conditions in which the individual circuits are to be exploited. For this reason, it is not possible to give precise values for general application.

Some typical values for this case are given in **Figure A10**, and are applicable to domestic consumers supplied at 230/400 V (3-phase 4-wires). In the case of consumers using electrical heat-storage units for space heating, a factor of 0.8 is recommended, regardless of the number of consumers.

Number of downstream consumers	Factor of simultaneity (ks)
2 to 4	1
5 to 9	0.78
10 to 14	0.63
15 to 19	0.53
20 to 24	0.49
25 to 29	0.46
30 to 34	0.44
35 to 39	0.42
40 to 49	0.41
50 and more	0.40

Fig. A10 : Simultaneity factors in an apartment block

4.5 Diversity factor

The term diversity factor, as defined in IEC standards, is identical to the factor of simultaneity (k_s) used in this guide, as described in 4.3. In some English-speaking countries however (at the time of writing) diversity factor is the inverse of k_s i.e. it is always ≥ 1 .

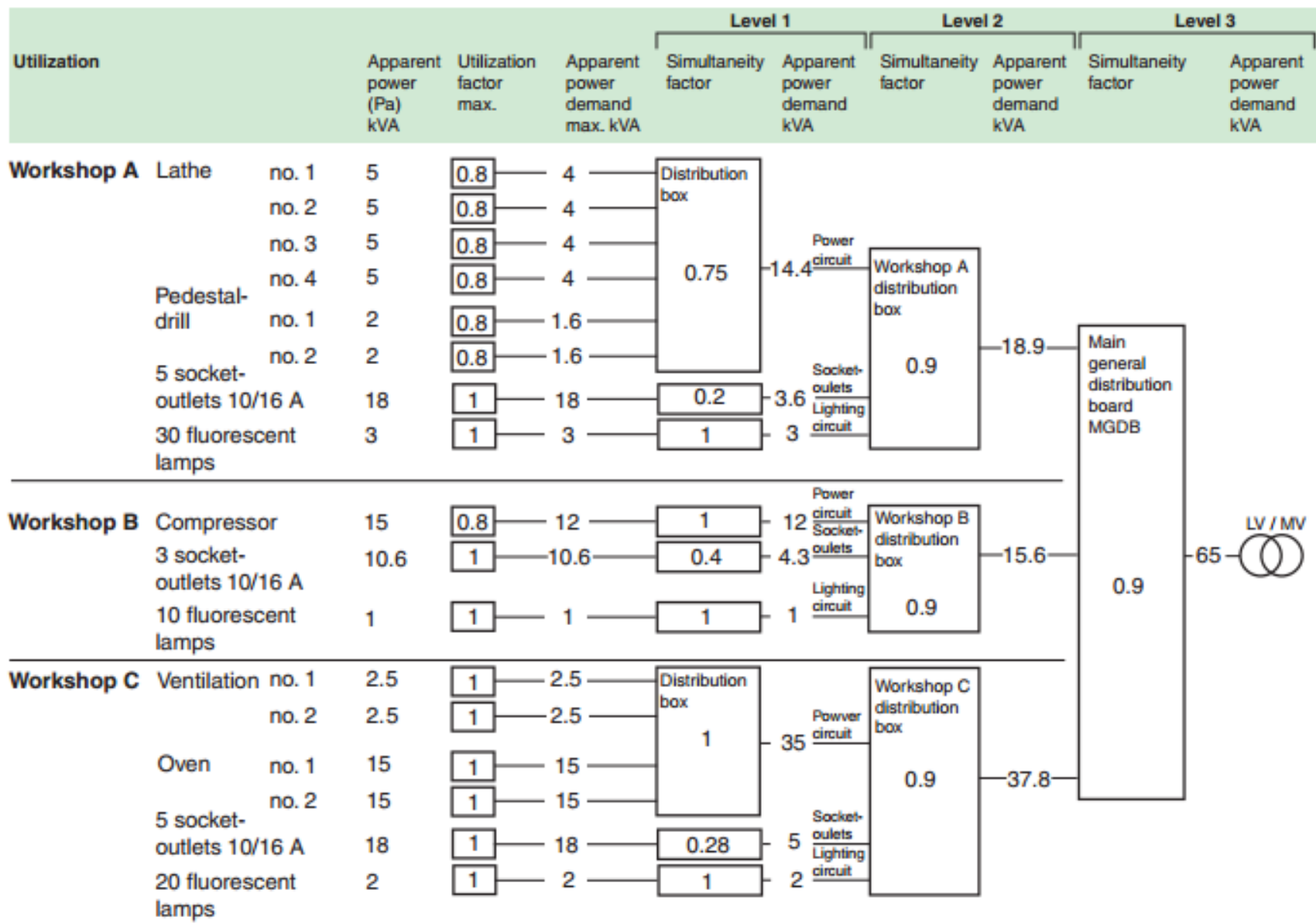


Fig A14 : An example in estimating the maximum predicted loading of an installation (the factor values used are for demonstration purposes only)